

HD189733b

R-0062

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HD189733_221022 · created 2026-07-15T14:57:08+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459874.6855 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1669 +/- 0.0093
Transit depth (Rp/Rs)^2	2.78 +/- 0.31 [%]
Orbital Inclination (inc)	86.24 +/- 0.75
Airmass coefficient 1 (a1)	1.001 +/- 0.014
Airmass coefficient 2 (a2)	-0.0013 +/- 0.0036
Scatter in the residuals of the lightcurve fit is	1.38 %
Best Comparison Star	#1 - [492, 165]
Optimal Aperture	5.66
Optimal Annulus	12.46
Transit Duration (day)	0.0786 +/- 0.0052

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1669 $\pm$ 0.0093 vs 0.15534 (deviation 0.96 σ)
Duration fit vs published	0.0786 d vs 0.0754 d (deviation 0.47 σ)
Mid-time O-C	2.9 min
Depth SNR	13.9
Residual scatter	1.38 %

Light curve

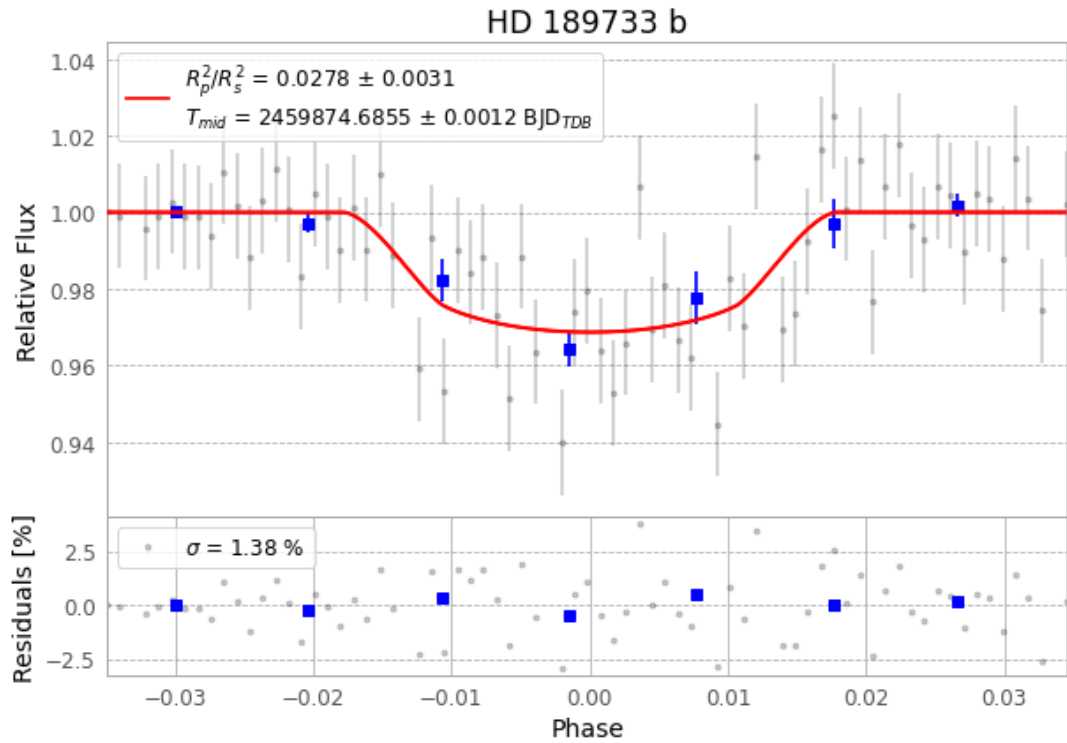


Figure 1 — FinalLightCurve_HD 189733 b_2022-10-21.png (SHA-256 54dca9d8a8561dab...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [526, 260], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9413c7ae14cae931...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

70 FITS frames (46 MB), HD189733221022023316.FITS → HD189733221022061515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HD189733-221022.log (141288e10faa...), FinalLightCurve_HD 189733 b_2022-10-21.png (54dca9d8a856...), FinalLightCurve_HD 189733 b_2022-10-21.csv (ccce80a26bea...)

Compute resources

Wall time 140.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.